MIL-S-20561A(OS)
14 March 1974
SUPERSEDING
MIL-S-20561
8 January 1952 and
Notice-1
19 November 1973

MILITARY SPECIFICATION

SPRING; ESCAPEMENT, CORROSION-RESISTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 The spring covered by this specification shall cover one class spring for mechanical timing movements, designated: spring, escapement, corrosion-resisting.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation - Packaging, Methods of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The springs furnished under this specification shall be a product which has been inspected and has passed the first article inspection specified herein (see 6.2).

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3.2 Material - Springs shall be made from steel produced by the crucible or electric-furnace process, using material of high quality.

3.3 Design and construction - The design and construction of springs shall conform to the requirements specified herein.

3.4 Chemical composition - Springs shall be made from steel containing not less than 25 percent nickel and not less than 3 percent chromium.

3.4.1 An analysis of each lot shall be furnished by the contractor, showing the percentage of each element specified.

3.5 Tensile - Finished springs shall be in the cold-rolled condition, and shall have a minimum tensile strength of 200,000 pounds per square inch and a maximum tensile strength of 250,000 pounds per square inch.

3.6 Ductility - Springs shall meet the ductility test described in paragraph 4.6.2.

3.7 Dimensions - Springs shall conform to the dimensions and permissible variations as specified in 6.2.

3.8 Workmanship - Springs shall be uniform in quality and temper and shall have a smooth, mirror-like surface finish, free from burrs, scale, and other injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 First article sample - Unless otherwise specified for each contract or order, a representative sample of five springs shall be submitted to the Government inspector, preserved, packaged, and packed as specified in the contract or order, for first article inspection at an activity designated in the contract or by the Naval Ordnance Systems Command. Further production of the springs by the contractor prior to approval of the contracting activity of first article samples shall be at the contractor's risk. First article samples accepted will be applied as part of the quantity specified in the contract or order.

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4.1.2 Inspection lot size - Unless otherwise specified, the number of units of production in an inspection lot will be as determined by the Government inspector.

4.2 Classification of inspection - The inspection and testing of the spring shall be classified as follows:

(a) First article inspection - First article inspection shall consist of examination and tests accomplished on samples which are representative of the production of the item after the award of contract, to determine that the first article samples meet the requirements of this specification.

(b) Quality conformance inspection - Quality conformance inspection shall consist of examination and tests performed on the spring manufactured and submitted for acceptance under contract.

4.3 First article inspection - The first article inspection of the spring shall consist of the visual examination (4.5) and the tests specified under paragraph headed Test Methods (4.6).

4.4 Lot - A lot shall consist of springs made from the same material and made from the same ingot of steel.

4.5 Visual examination - All springs shall be examined by the inspector.

4.6 Test methods -

4.6.1 Tensile - At least five speciments each 10 inches long and cut 10 feet from the ends of the finished flattened wire, and five specimens from approximately the middle of the flattened wire, shall be submitted by manufacturer for tensile tests. When tests are to be made at the point of delivery, these speciments shall accompany each shipment of springs constituting an individual lot.

4.6.2 Ductility - At least five springs shall be taken from each lot and shall be bent cold through an angle of 180 degrees around a rod having a diameter two times the thickness of the spring material without showing cracks or fracture on the outside surface.

4.6.3 Dimensional variations - At least 25 springs shall be selected from each lot for dimensional measurements. The thickness of the springs shall be measured on a precision measuring machine approved by the Bureau of Standards.

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4.6.4 Rejection - Springs in which injurious defects are revealed subsequent to acceptance and which fail to comply with the requirements of the contract or purchase order shall be rejected.

5 PREPARATION FOR DELIVERY

5.1 Preservation and packing - Preservation and packaging shall be in accordance with MIL-P-116 Method III. Springs shall be carefully packed in a glass or suitable transparent plastic tube. No more than 1000 springs shall be packed in each container.

5.2 Marking - Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The spring covered by this specification is intended to be used for mechanical timing movements for escapement lever.

6.2 Ordering data - Procurement documents should specify the following.

(a) Title, number, and date of this specification

(b) Dimensions and tolerances required

(c) Quantity required

(d) When a first article sample is not required in contract or order (see 3.1)

(e) Level of preservation and packing required if other than 5.1

Preparing activity:
Navy - OS
Project No. 5360-N002

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